
Automating the Bill Review Process

Right now, someone compares every vendor bill to the quote by hand, then types it into QuickBooks a second time. This is a plan to have a computer do the comparing and the typing — while people keep making every decision that involves money.

WHO THIS IS FOR

This is the plain-English version. No technical background needed. It is meant to be read by anyone who has to understand, approve, use, or help set up this system — including our IT company. A separate technical document exists for whoever builds it.

PART ONE

The problem we are trying to solve

Four things that happen today, every week.

Here is how buying something for a job works today. We ask vendors for **quotes**. We pick one and place the order — sometimes with a formal purchase order, often with just an email saying "go ahead." Weeks later, the vendor's **bill** shows up. Then someone has to check it.

1 Checking the bill against the quote is done by eye

Someone opens the quote and the bill side by side and reads down both lists, line by line, looking for anything that doesn't match. On a bill with forty line items, that is forty comparisons. It is slow, it is boring, and boring is exactly when people miss things.

2 When something is wrong, it is usually small and easy to miss

A price quoted at \$4.10 arrives at \$5.85. A quantity is billed higher than what was delivered. Freight and handling appear that were supposed to be included. Any one of those is easy to overlook. Across hundreds of bills a year, they add up to real money that we simply paid.

3 Everything gets typed in twice

After the bill is checked, someone types the vendor, the amount, the date, the job, and the account codes into QuickBooks — copying information off a document they just finished reading. That is the same work done twice, and each retype is another chance for a typo.

4 Nobody has a reliable picture of the week ahead

"What do we owe over the next seven days, and what's coming in to cover it?" is a question that requires someone to go and assemble the answer. So it gets asked less often than it should.

THE SHORT VERSION

We are paying skilled people to do comparison work that a computer is better at, and the comparison still isn't reliable — because it depends on a person being alert on the fortieth line of the ninth bill that day.

PART TWO

What we want to build

Three things, in the order they'd be built.

FIRST Catch the bills that don't match The computer reads every quote and every bill, compares them line by line, and flags anything that doesn't line up — with the dollar amount attached.	SECOND Put approved bills into QuickBooks Once a person has reviewed and approved a bill on screen, the system enters it into QuickBooks for them. No retyping.	THIRD A seven-day money report Every Monday morning: what has to go out this week, what should come in, and where that leaves us. Sent automatically.
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What each one actually changes

	TODAY	AFTER
Checking bills	Read every line of every bill and hope you catch it.	Read only the bills the system flagged — typically a small fraction — and it tells you which line and why.
Entering bills	Type the whole thing into QuickBooks by hand.	Check what the system read off the bill, correct anything wrong, click approve. It does the entry.
Cash view	Someone builds it when asked.	In your inbox Monday morning, and on screen any time.
Vendor overcharges	Caught sometimes. Nobody knows the real rate.	Caught consistently, with a record of which vendors do it repeatedly.

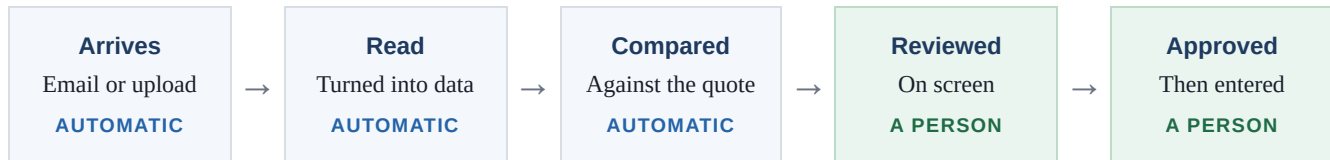
ONE THING TO BE CLEAR ABOUT

This does **not** pay anybody, and it does not decide anything. It reads, compares, flags, and — only after a person approves — types. Every decision that involves money stays with a person. That is a deliberate design choice, not a limitation we plan to remove later.

PART THREE

One bill, all the way through

This is the whole system. If you follow this page, you understand it.



Example below uses made-up names and figures purely to illustrate the steps.

1

AUTOMATIC

The bill arrives

Baker Supply emails an invoice for job 4417 to our accounts-payable inbox. The system picks it up within five minutes. Nobody forwards it, files it, or does anything. If a bill comes in some other way — handed over, faxed, downloaded — anyone can drag it onto the screen instead.

2

AUTOMATIC

It gets read

The system reads the PDF and pulls out the vendor, invoice number, date, due date, every line item, quantity, unit price, tax, freight, and the total. It works on scanned and photographed documents too, not just clean PDFs. It also checks straight away whether we have seen this invoice before — same file, or same vendor and invoice number. **Vendors resend invoices, and paying one twice is real money out the door.**

3

AUTOMATIC

It gets compared to what we agreed to

The system finds the matching quote — by purchase order number, by quote number, or by vendor and job — lines the two documents up item by item, and checks every number. It also re-adds the bill to confirm the vendor's own arithmetic is right.

In our example it finds two problems: the copper fitting was quoted at \$4.10 and billed at \$5.85 across 400 units, and \$180 of freight was added that the quote showed as included. **Total overcharge: \$880.**

EVERYTHING SO FAR HAPPENED WITHOUT ANYONE TOUCHING IT

Steps 1 to 3 run on their own, within minutes of the bill landing. A person's first involvement is the next step — and by then the work is already laid out for them.

One bill, all the way through

Where a person picks it up.

4

A PERSON

Someone reviews it

The bill appears in a review list. On one screen: the actual PDF on the left, and what the system read from it on the right, with the two problems highlighted and explained in a sentence. Every number links back to the exact spot on the page it came from, so nothing has to be taken on faith.

The reviewer can correct anything the system got wrong, and either accept the charge with a note, or hold the bill and query it with the vendor.

5

A PERSON

They approve it, and it goes into QuickBooks

One click. The system enters the bill in QuickBooks with the vendor, the lines, the job, the account codes, and the due date already filled in, and links it to the purchase order if there is one. Then it reads the entry back to confirm it went in correctly.

Nobody retypes anything. The paperwork stays attached to the record on our side, one click away from the QuickBooks entry.

What that changes for the person doing it

FOR ONE ORDINARY BILL	TODAY	AFTER
Find the matching quote	Search email and folders	Already attached
Compare the lines	Read both documents in full	Read the flagged lines only
Enter it in QuickBooks	Type it all in	Check and click approve
What the person is actually doing	Copying and comparing	Judging the exceptions

THE POINT OF THE WHOLE THING

The work that disappears is the copying and the line-by-line reading — the part that is tiring, repetitive, and unreliable when done by a person. The work that stays is the part that needs judgement: **is this charge acceptable, and do we pay it?** Nobody is removed from the process. The boring half of their job is.

PART FOUR

What the computer does, and what it never does

This is the part people worry about, and it is worth being precise.

THE SIMPLEST WAY TO THINK ABOUT IT

Think of it as an extremely fast, extremely literal clerk who reads every page and never gets bored, never skims, and never has a bad afternoon — **but who is not allowed to do the arithmetic and is not allowed to decide anything.**

It reads. A calculator does the maths. A person decides. Those three jobs are kept separate on purpose, and that separation is what makes the system safe to trust.

THE COMPUTER DOES THIS

A PERSON ALWAYS DOES THIS

Reads the document and pulls out the numbers	Confirms the numbers are right, with the original on screen beside them
Compares the bill to the quote, line by line	Decides whether a difference is acceptable or gets challenged
Adds everything up and flags what doesn't match	Approves the bill — nothing reaches QuickBooks without this
Enters the approved bill into QuickBooks	Decides what actually gets paid, and when
Drafts the email querying an overcharge	Reads it, changes it if needed, and sends it

The arithmetic point, because it matters

The part that reads documents is artificial intelligence, and AI can make mistakes. So it is used only for reading — turning a picture of an invoice into a list of numbers. **Every comparison and calculation afterwards is ordinary accounting arithmetic**, the way a spreadsheet does it: same inputs, same answer, every time.

The AI is never the thing that decides whether two numbers are different — and that distinction is what separates a tool a finance department can rely on from one it can't.

Three possible outcomes for any bill

CLEAN

Everything matches the quote. Goes to review as a quick confirmation, not an investigation.

WORTH A LOOK

Something differs but within reason — a small price move, an extra line. Flagged with the amount.

STOP

Duplicate invoice, the maths doesn't add up, or billed for more than was ordered. **Cannot be approved until a person resolves it.**

ON TOLERANCES

We set what counts as "worth a look" per vendor — a 2% move on steel is market noise, a 2% move on a fixed-price subcontract is not. Those thresholds are ours to change, and they are written down rather than living in someone's head.

The QuickBooks complication

Why this needs two computers instead of one.

There are two versions of QuickBooks, and which one you have changes how hard this is. **We have QuickBooks Desktop** — the one installed on a computer, with the company file on our server.

THE ANALOGY

QuickBooks Online is like a phone number. Any authorised system can call it from anywhere and ask it to do something.

QuickBooks Desktop is like a filing cabinet in a locked office. You cannot phone a filing cabinet. Somebody has to physically be in that room to open the drawer. That "somebody" is the second computer.

So the system is in two pieces. This is not a design choice we made; it is a condition QuickBooks Desktop imposes.

1 · The main system

Runs everything you actually see and use.

- Reads the emails and documents
- Does the comparing
- Hosts the review screen staff log into
- Sends the Monday report

Can live at our IT company or with a hosting provider.
Nobody has to be near it.

2 · The QuickBooks messenger

A small Windows computer that sits next to the QuickBooks file.

- Its only job is passing messages
- Fetches bills and invoices out
- Puts approved bills in
- Nobody logs into it or uses it

Must be on our network with the QuickBooks file, and must stay switched on.

Two practical consequences, stated plainly

It is not instant. The messenger checks in on a schedule rather than reacting the moment something happens. In practice that means minutes, not seconds. For accounts payable, that is fine.

If we ever move to QuickBooks Online, that second computer disappears and this gets simpler. The system will be built so switching is a change to one component, not starting over. Worth knowing, because Intuit has been steadily pushing customers toward the online version.

THE ONE THING TO CHECK FIRST

Our IT company has to be willing to run that second computer — a small Windows machine that stays permanently switched on and logged in, running software that talks to the QuickBooks file. Most will. Some have policies against it. **That conversation should happen before anything is built**, because a "no" changes the plan.

PART SIX

Who has to run it day to day

Nobody. That is the honest answer, and here is why.

A fair question about anything automatic is: what happens when the person who set it up is on holiday? The answer is that **there is no daily start-up step**. Setting it up happens once. After that it runs the way a phone system or an email server runs — continuously, without anyone starting it.

TASK	WHO	HOW OFTEN
Set up the two computers	IT company	Once , at the start
Check the email inbox, read documents, compare bills, sync QuickBooks, send the Monday report	The system itself	Continuously, on its own schedule
Start back up after a power cut or a restart	Automatic	Whenever needed, unattended
Review flagged bills and approve them	Accounts payable	As bills arrive — <i>this is the job they already do</i>
Respond if an alert says something broke	One named person	Rarely
Updates and password rotation	IT company	Quarterly

How it restarts itself

Both computers are configured to bring everything back automatically after a restart — including after a Windows update reboots the messenger machine overnight, which it will. The timetable for recurring work lives inside the software, not in someone's calendar. It runs whether anyone is logged in, at their desk, or away.

The alarm that actually matters

The real risk with anything that runs quietly in the background is not that it crashes loudly — it is that it stops and **nobody notices for a week**. A system that has quietly stopped looks exactly like one with nothing to do.

So it watches itself: if QuickBooks hasn't been checked in hours, or the inbox hasn't been read, or an approved bill is sitting unsent, it sends an alert to a person by email. **That alarm is what replaces someone checking on it.**

ONE THING WE DO NEED

Name one person the alerts go to. Not to run it — just so a problem has somewhere to land, and a stalled system isn't discovered by the controller wondering where Monday's report went.

PART SEVEN

How long, and what it costs

Estimates. The first two weeks exist specifically to firm these up.

Roughly three months to all three pieces working

Each stage is usable on its own, so value arrives before the end rather than all at once.

WEEKS 1–2 Prove it works
Confirm we can read from and write to our QuickBooks. Gather 30–50 real quote-and-bill pairs and measure how accurately the system reads them. **If something here doesn't work, we find out now — before spending money on the rest.**

WEEKS 3–6 Catching mismatched bills
The reading, comparing, and review screen. Runs *alongside* the current manual process, not instead of it, so we can measure whether it agrees with our people before anyone relies on it.

WEEKS 7–10 Entering bills into QuickBooks
The approve-and-enter step. This is the one accounts payable will feel most, because it removes the retyping.

WEEKS 11–12 The seven-day money report
Automatic Monday report and a live version on screen any time.

Ongoing running cost

WHAT	ROUGH MONTHLY	NOTE
The AI that reads documents	\$40–60	Charged per document read. At our volume this is small.
Hosting the main system	\$50–100	May be less if our IT company hosts it on the server they already run.
QuickBooks Desktop connection	\$200–500	The number to confirm. Buying this saves roughly a month of build work.
Estimated total	\$300–650	To be confirmed with vendors in the first two weeks.

WORTH PUTTING NEXT TO THAT NUMBER

A single missed overcharge of the size in the earlier example — \$880 on one invoice — is more than a month of running the whole system. The question worth asking is not whether it pays for itself, but **how often this is happening now without anyone knowing**. Nobody can currently answer that, which is itself part of the problem.

PART EIGHT

What could go wrong

The fair objections, and the honest answer to each.

THE WORRY	WHAT WE DO ABOUT IT
"It reads a number wrong and we pay the wrong amount."	A person sees the original document beside every number before approving, and each figure links to where it came from on the page. The bill's own totals are re-added independently — if they don't agree, it cannot be approved.
"It enters the same bill twice."	Duplicate checking happens before anything else, and each entry carries a unique marker so a retry after a dropped connection can't create a second copy.
"Our IT company won't allow the second computer."	We ask them in week one, before anything is built. There is a fallback that uses outside hosting instead.
"Staff won't trust it and will keep checking everything anyway."	It runs alongside the manual process first, and we publish how often it agreed, what it caught that a person missed, and what it missed. Trust comes from those numbers, not from a demo.
"It stops working and nobody notices."	It alerts a named person when it hasn't done its work — the alarm is on <i>silence</i> , not just on errors.
"Half our orders have no purchase order, so there's nothing to compare to."	The system treats the "go ahead and order it" email as the record of what was agreed. That also finally answers "did anyone actually approve this order?" — which today is a memory test.
"An auditor asks why something was approved and we can't explain it."	Every action is permanently recorded — what was read, what was flagged, who approved it, when, and what they changed. That is a better trail than we have today, where the reasoning lived in someone's head.
"Vendor bank details and tax numbers are on these documents."	Documents are stored encrypted, access is limited to the people who need it, and nothing sensitive is written into logs. Worth reviewing with our IT company as part of setup.
"What if we hate it and want to stop?"	QuickBooks stays the official record throughout — this system feeds it, never replaces it. Switch it off and you are back to the current process with nothing lost and nothing to unpick.

THE RISK NOBODY LISTS

Doing nothing is not the safe option. Today we do not know how much we are overpaying, because nothing measures it. That number is not zero, and it stays invisible for as long as the checking is done by eye.

What we need from you

Three decisions, and one thing that can start today.

1 Will our IT company host this?

Specifically: a small Windows computer that stays on and logged in next to the QuickBooks file, plus space for the main system. **Ask them this week** — their answer shapes everything else, and there is a workable fallback either way.

2 Do we buy the QuickBooks connection or build it?

Companies sell a ready-made connection to QuickBooks Desktop for a few hundred a month. Building our own saves that fee but adds roughly a month of work and leaves us maintaining it forever. **The recommendation is to buy it** — it is not the part of this that makes us money, and it is the part most likely to break on a QuickBooks upgrade.

3 Who owns this?

One named person who gets the alerts and makes the calls on thresholds. Not a daily job — but "everyone" and "nobody" are the same answer in practice.

What to actually ask the IT company

Four questions. Their answers decide the shape of everything else, so this call is worth making before any other step.

- Where does our QuickBooks company file physically live, and which version is it?
- Can you run a small Windows machine that stays switched on and logged in, next to that file, running software that reads from and writes to QuickBooks? At what cost?
- Can you also host the main system, or should we host it elsewhere?
- Do you have any policy against third-party software connecting to QuickBooks?

If the answer to the second question is no, that is fine — it means hosting the main system outside and finding another home for the small Windows machine. Better to know in week one than in week eight.

ONE THING THAT CAN START IMMEDIATELY

Collecting **30–50 real quote-and-bill pairs** — including the messy ones and the ones with no purchase order — needs no decisions, no budget, and no computers. It is how we measure whether any of this works, and it is on the critical path. **It could start this afternoon.**